

TerraTwin

AI Readiness Digital Twin for Sustainable Agriculture

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1. Introduction

Agricultural AI often arrives as a finished recommendation—irrigate, fertilise, spray or plant—before the farm has the data, connectivity, governance practice or safe trial environment needed to use that recommendation responsibly. TerraTwin is a browser-based AI-readiness digital twin designed for smallholders in Jazan, Saudi Arabia. It measures readiness, rehearses agricultural decisions in a sandbox, and then supports daily farm operations so readiness improves through ordinary use rather than a one-time survey.

The solution implements the ITU-T Y.3172 ML pipeline as visible application behaviour: $SRC \rightarrow C \rightarrow PP \rightarrow M \rightarrow P \rightarrow D \rightarrow SINK$, orchestrated within a sandbox. The current Model node is explicitly prototype-grade and rule-based, not a trained statistical or deep-learning model; all farm figures are illustrative. This disclosure is intentional so the prototype demonstrates architecture, policy enforcement and readiness logic without overstating model accuracy.

Why Jazan

Jazan contains sharply different agricultural systems within one region: the hot humid Tihama coast, Wadi Sabya, Abu Arish plain, Al-Reeth foothills and Fifa highlands. A recommendation useful for one setting may be harmful in another. TerraTwin therefore chooses site before crop and makes local conditions part of every simulation and guidance path.

Core solution

- Readiness scorecard: 19 questions across Data, Infrastructure, Governance, Skills and Sandbox Readiness, scored out of 100 with the weakest dimension named as the priority gap.
- Decision sandbox: weekly decisions—irrigate, fertilise, treat, inspect or hold—change six live crop vitals and pass through the full Y.3172 pipeline; the Policy node can visibly refuse unsafe actions.
- Operations layer: plant register, irrigation/fertilisation scheduling, plant-health camera with offline fallback, weather-to-action alerts, farmers' forum, local market and Farmy knowledge assistant.

2. Use Case, Existing Gaps and Value

2.1 Gaps in existing approaches

Gap	Why it matters	TerraTwin response
Farm-level readiness	National or regional readiness indicators do not reveal whether an individual holding lacks data, connectivity, skills, governance or a safe testing route.	Five-dimension score with a named priority gap enables targeted support before investment.
Unsafe first deployment	AI advice can be acted upon before it is validated against local agronomic or governance constraints.	Every candidate decision is simulated first; Policy rules can block it before it reaches the user.
Weak operational data	A survey captures readiness at one moment but does not create the farm history needed for future AI.	Each logged irrigation, scan and schedule change becomes a farm-owned operating record.
Connectivity barrier	Cloud-only tools can fail where the field signal is weakest.	Core app works locally; photo analysis and weather features degrade to documented offline fallbacks.
Local-context mismatch	Crop suitability and water decisions vary strongly between coast, plains, foothills and highlands.	Simulation uses Jazan sub-region, soil, season and weather context before generating outcomes.

2.2 Practical value

- Failure without cost: a wrong irrigation or fertiliser decision loses a simulated week or season instead of water, inputs and yield.
- Governance made executable: privacy and safe-practice rules are represented in the P (Policy) node and logged as part of the decision path.
- Readiness that maintains itself: the heaviest readiness requirement—usable farm data—improves as the farmer uses the scheduling, scan and simulation features.
- Water efficiency at the decision point: rain-aware rescheduling can skip unnecessary irrigation and count estimated litres saved where behaviour changes.
- Inclusive interaction: the app is designed for phone use and supports voice notes in the community layer, reducing dependence on typing.

2.3 AI Readiness alignment

The project follows the hackathon framing of AI readiness as more than model performance. TerraTwin's operational five dimensions are a farm-level implementation lens: Data and Infrastructure address foundational capacity; Skills addresses adoption capability; Governance addresses consent, privacy and safe use; Sandbox Readiness addresses controlled experimentation, validation and human oversight. The solution is also aligned to national AI strategy, water-efficiency priorities and farm-level deployment realities through the policy mapping below.

3. Mapped Documents and Y.3172 Pipeline

For each Y.3172 node, TerraTwin identifies both the technical role and the policy/strategy materials that support responsible implementation. The complete submission knowledge base contains 505 authentic public sources; this table lists the highest-priority sources used directly by the prototype.

Node	TerraTwin implementation	Mapped documents / policy rationale
SRC	Farm profile, soil, irrigation, weather, prior yield and pest observations.	PDPL (SDAIA); MEWA agriculture/water guidance; GASTAT agricultural statistics; FAO/AQUASTAT.
C	Consolidates observations into a farm profile and local operational record.	PDPL purpose limitation/consent; SDAIA data governance; ITU-T Y.3172 data-flow role.
PP	Normalises inputs into readiness and simulation formats.	ITU-T Y.3172; data-quality and governance guidance represented in KB; transparent transformation rules.
M	Rule-based readiness scorer and digital-twin growth/stress/pest engine; replaceable later by trained model.	ITU-T Y.3172 architecture; ITU AI Readiness reports; future validation benchmark references in KB.
P	Checks irrigation interval, fertiliser ceiling, pest threshold and data-handling flags; can refuse an action.	SDAIA AI governance; Saudi PDPL; MEWA safe/resource-efficient practice; human override/guardrails.
D	Routes approved output to dashboard, alerts, recommendation report and logs.	Data-minimisation/access principles; usability and inclusion; energy/connectivity-aware delivery.
SINK	Farmer-facing action/result; the farmer remains decision-maker.	Human-in-the-loop principle; clear disclosure that prototype outputs are decision support, not certified agronomic diagnosis.
MLFO / Sandbox	Sequences PP→M→P→D and rehearses decisions before real-field use.	ITU-T Y.3172 pipeline and sandbox concepts; controlled testing before deployment; traceable run history.

Priority reference set

ITU-T Y.3172 (ML pipeline); ITU AI Readiness report v2.0 and AI-RE Toolkit; Saudi National Strategy for Data and AI (SDAIA); SDAIA Regulations and Policies / AI governance; Saudi Personal Data Protection Law; Ministry of Environment, Water and Agriculture; Saudi Vision 2030; GASTAT agricultural statistics; FAO and AQUASTAT; ICBA and ICARDA; UNESCO Khawlan coffee heritage sources. Full clickable references are supplied in the separate Knowledge Base PDF/DOCX.

4. Evaluation Scenarios

The scenarios test not only successful operation but also controversy and failure. Each response is tied to pre-agreed rules so corrective action is not improvised after harm occurs.

Scenario	Risk	System response	Policy/KB basis
1. Over-irrigation request	A farmer irrigates again while soil is already saturated or before the minimum safe interval.	P node refuses the request, turns visibly to blocked status, states the reason, records the event and leaves plant state unchanged. Human can inspect and choose a different action.	Water-efficiency / safe-practice rules; Y.3172 Policy node; MEWA-aligned resource stewardship.
2. Excess fertiliser	The farmer requests another dose after the configured seasonal ceiling.	Policy node blocks the dose and explains runoff/root-stress risk. The simulator demonstrates the avoided consequence without consuming input.	Safety guardrail; environmental protection; transparent override conditions.
3. Wrong or uncertain AI diagnosis	Gemini or the offline image analysis returns a low-confidence or incorrect plant-health assessment.	Output is presented as decision support, not certified diagnosis; confidence/mode is surfaced; user is directed to inspect, consult the community or seek expert confirmation before treatment. Production deployment requires expert-labelled validation.	Human-in-the-loop; model validation; risk disclosure; agricultural extension escalation.
4. Connectivity/API outage	The farm loses internet during diagnosis, weather retrieval or assistant use.	Core app continues locally. Camera analysis falls back to on-device colour analysis; weather uses documented offline climate logic; the interface states which mode is active.	Resilience, availability and inclusion; no silent degradation.
5. Data misuse / targeted advertising controversy	A future commercial version attempts to use farm records, diagnosis results or inferred vulnerabilities for unrelated advertising or resale.	Purpose-limitation and consent rules trigger. Data use is blocked unless separately lawful and consented; access is logged; offending processing is disabled and referred for governance review. Prototype keeps data local and collects no advertising profile.	Saudi PDPL; SDAIA governance; purpose limitation, consent, minimisation and accountability.
6. Regional crop mismatch	A crop is selected for a location whose heat/chill/soil profile is unsuitable.	Site is selected before seed; simulator returns a suitability warning and allows a safe simulated trial rather than field commitment.	Climate adaptation; evidence-based extension; local-context validation.

5. Submission Contribution, Limitations and Deployment Plan

5.1 Knowledge-base contribution

- 505 authentic public sources: 83 AI-readiness/policy/standards sources and 422 agricultural/technical sources, with reference numbering preserved for traceability.
- Reusable agriculture AI-readiness use-case pattern expressed in Y.3172 pipeline terms.
- Agriculture-specific readiness rubric, crop/region climate-matching method, weather-to-action rules, plant-health prompt contract and Jazan crop scheduling defaults.
- Policy-node safeguard pattern that checks candidate recommendations before they are surfaced to a farmer.

5.2 Prototype limitations and disclosures

- The M node is rule-based; no trained statistical/deep-learning model or production accuracy claim is made.
- All farm data is illustrative. Authentication and localStorage are prototype-grade and must be replaced by a secure backend/identity layer for production.
- Vision diagnosis has not been validated against expert plant-pathologist labels for Jazan crops; field deployment requires such validation.
- Online weather and Gemini services are optional dependencies; offline modes are intentionally coarser and labelled as such.
- Community content is device-local in the prototype; regional deployment requires server-side identity, moderation and governance.

5.3 Next deployment steps

Phase	Action	Evidence of readiness
1. Validate	Pilot with agronomists and a small number of Jazan farms; validate crop rules and vision outputs.	Error analysis, expert agreement, documented failure modes.
2. Secure	Replace localStorage with consent-aware backend, access control, audit logs and deletion/export controls.	PDPL-aligned data map and security review.
3. Learn	Use consented farm records to train/validate a predictive M node while preserving surrounding Y.3172 nodes.	Versioned model card, validation report, rollback path.
4. Scale	Arabic RTL interface, wider crop coverage, extension/MEWA integration and regional dashboards.	Field adoption metrics, water/decision outcomes, readiness-score improvement.

5.4 Submission files

Technical report (this document); demo video (≤ 7 minutes); GitHub-ready source code; 505-source Knowledge Base; AI Readiness Scorecard workbook; supporting pitch deck and extended project explanation. The final submission package groups all deliverables, including the completed demo video and the confirmed contact email.

Project statement: TerraTwin tests the decision before the crop. It measures whether a farm is ready for AI, provides a safe place to be wrong, and makes governance visible inside the same pipeline that produces the recommendation.